CSE490 (D)

INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERAITON

SPRING 2026

- Professional Certificate from BFF
- Internship Opportunity at BFF Data Science Department after successful completion
- Learning from the industry experts of Football Data Science
- Opportunity to co-author research papers for Sports Data Conferences

CSE490 (D)
Introduction to Football Data & Analytics
Lecture-2

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Motivation

Why are we analyzing football data? Are we gonna give advice to Pep Guardiola?

Motivation

We want to provide another way of seeing things. How?

Take data $\xrightarrow{\text{Convey}}$ ***Powerful Idea***

$\Big\downarrow$ How??

Pictures

Passing Networks

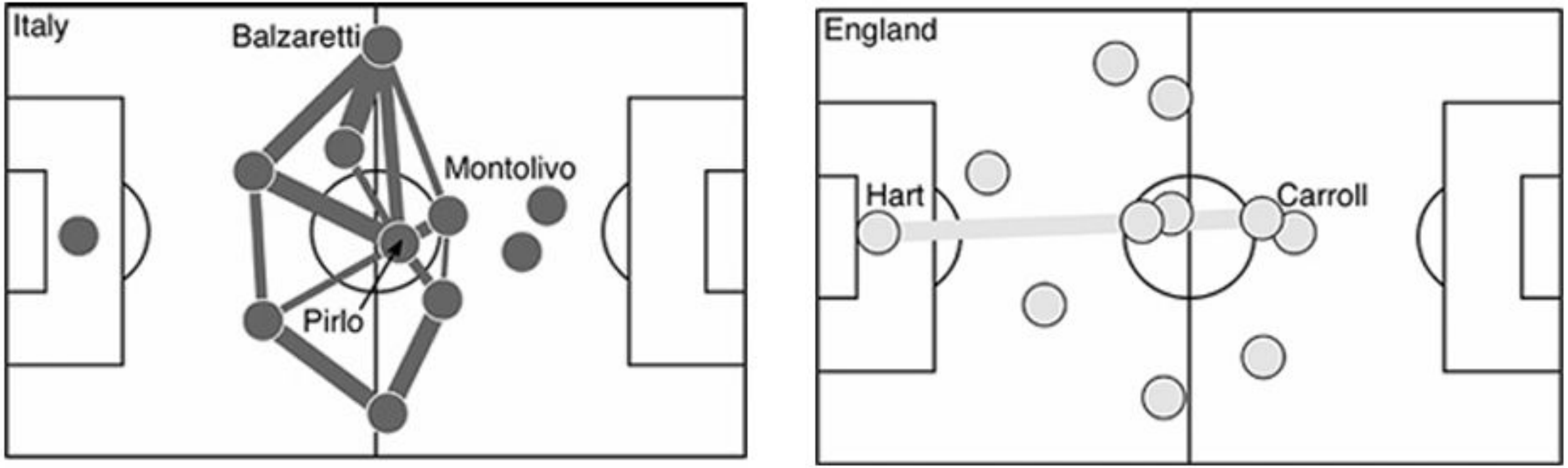

Figure 7.1 Passing networks for Italy (top) and England (bottom) in the Euro 2012 quarter-finals, showing links between pairs of players that completed 13 or more passes between them.Thicker lines indicate a higher number of passes. Both teams are attacking left to right.

What are the strategies the teams adopted? What are the issues with those strategies?

Passing Network

- ***Italy***⟶ Passing ⟶ Center: Pirlo
- ***England***⟶ Long ball ⟶ Target : Caroll **(Tall)**

➔ Problem with England's strategy? – No passing Network!

Result? – **The match went to tiebreaker!!!**

➔ Italy was no better! Too centralized. Centrality= 19.7%

Read: Soccermatics Chapter 7

Notebook:

https://colab.research.google.com/drive/1qf4PY7QJuZD6cM-NttZNMfJ_DLIVYH2T?usp=sharing

Measuring Centrality (Thomas Grund)

- ➔ Count the total passes each player **made + received**.
- ➔ Identify the player with the **highest total involvement** (most passes).
- ➔ For every other player, calculate the **difference** between the maximum value and their value.
- ➔ Take the **average of these differences**.
- ➔ Divide this average by the **total number of passes in the team**.

The result is a number between **0 and 1**:

- **0** → passes are evenly distributed (low centrality).
- **1** → one player dominates all passes (high centrality).

Measuring Centrality

Suppose a team has 4 players with the following total involvements:

- Player A = 80 passes, Player B = 50 passes, Player C = 40 passes, Player D = 30 passes

Step 1: Maximum involvement = 80

Step 2: Differences from maximum

A: 80 − 80 = 0, B: 80 − 50 = 30, C: 80 − 40 = 40, D: 80 − 30 = 50

Step 3: Average difference

(0+30+40+50)/4=30

Step 4: Total team passes = 200

Centrality measure: 30/200=0.15

Measuring Centrality

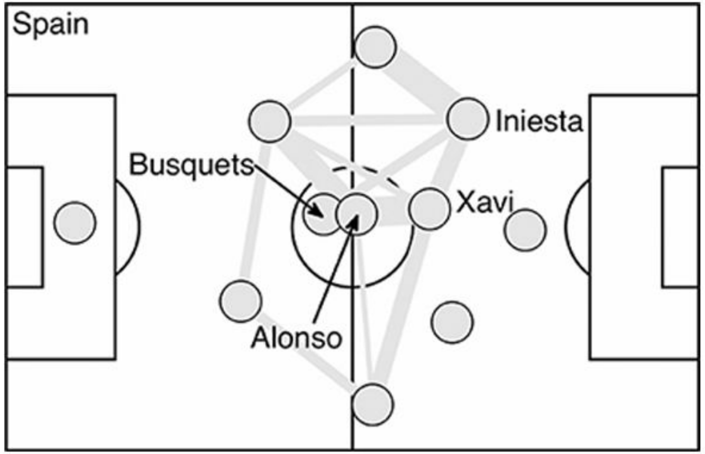

Figure 7.2 Passing network for Spain playing against Croatia in Euro 2012, showing links between pairs of players that completed 11 or more passes between them.Thicker lines indicate a higher number of passes.

Being decentralised gives a team an 8% advantage in terms of scoring rate.

Like Italy, Spain had most of the possession, with control of the ball for 71% of the match. They also had a similar **pass rate**, 9.65 passes per minute while in possession, compared with Italy's 9.15 times per minute against England.The calculated **centrality** measure for **Spain** is **14.6%**, compared to **19.7%** for Italy. **Scoreline: 1-0!**

Pass Heat maps

- Heat maps are spatial histograms.
- Each square on the heat map shows the number of passes made by the player from that square.
- The ‘hotter’ the square, the more passes made.
- We can count all passes for the heat map, but that will tell us little about a team’s play.
- To add context, we can set criteria such as passes that lead to shot in less than 15 seconds.

Pass Heat maps

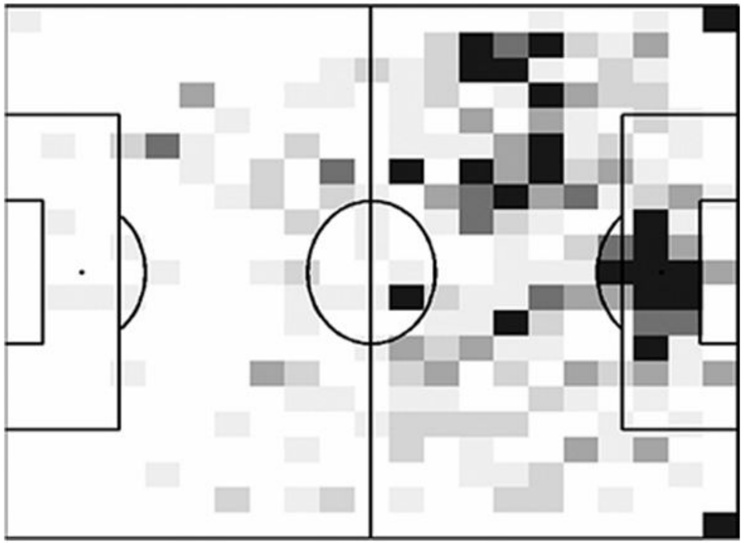

Figure 7.12 Real Madrid danger-zones during the Champions League season 2014/15.The shading is proportionally darker in areas where the ball was located during the 15 seconds leading up to a shot from the 20m by 20m area in front of the opposition goal. Data provided by Opta.

- **Darker areas = higher shot risk** within 15 seconds; lighter areas = lower risk.
- **Corners & inside the box** are naturally high-risk zones.
- **Left side outside the box** is a key hotspot (Marcelo & Ronaldo), where many dangerous chances originate.
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Comparing Players Using Heatmap

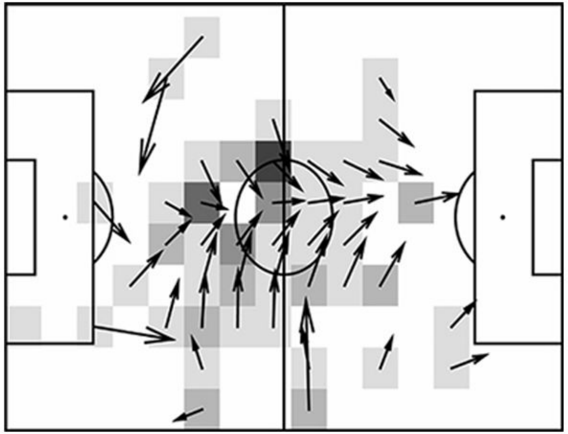


Figure 3.7 Pirlo's passing during Italy's semi-final against Germany in Euro 2012. The intensity of the shading at each particular position on the pitch indicates how often Pirlo made a pass from this position.The arrows are a statistical fit to all 66 passes he made during the match. The average direction of passes is indicated by the direction of the arrow. The average length of the arrow is proportional to the average length of a pass.

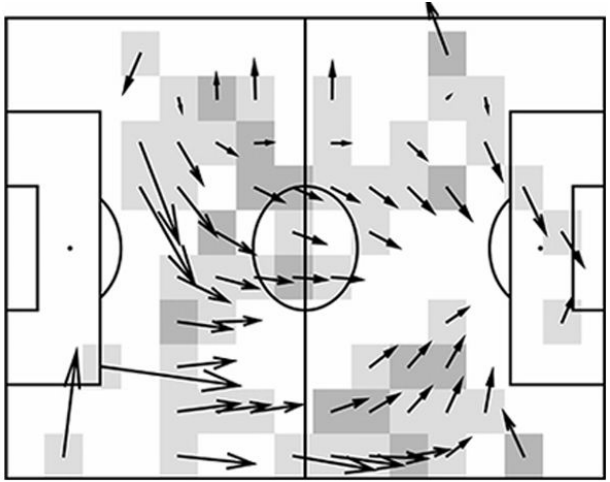


Figure 3.8 Schweinsteiger's passing during Germany's semi-final against Italy in Euro 2012. See Figure 3.7 for details of how this plot was created.

Read: Soccermatics Chapter 3, Picturing Pirlo

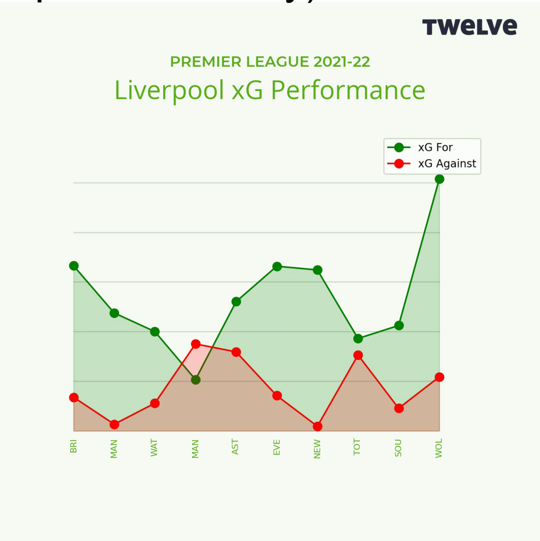
Key Performance Indices (KPI)

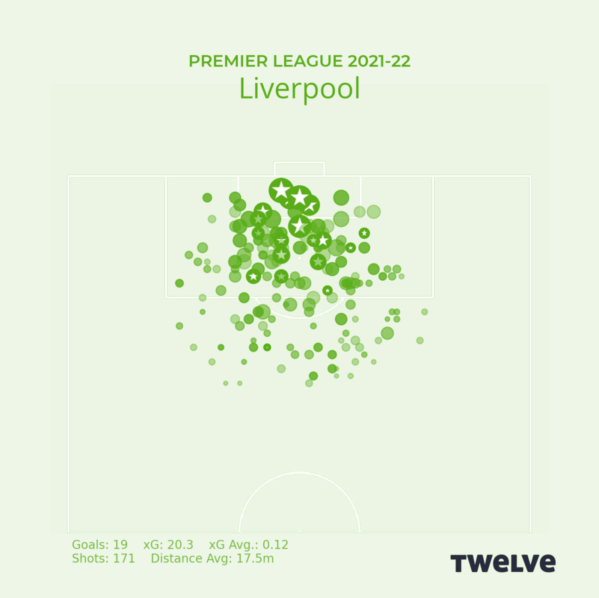
- Clubs should build a KPI-driven culture aligned with their playing style.
- KPIs set collaboratively by coaches, data scientists, and management.
- Measuring performance over 6–10 matches gives a reliable view of form.
- Example: Liverpool’s last 10 matches (Premier League 2021–22).

Key Performance Indices (KPI)

Expected Goals (xG)

- Shot map: Circle size = chance quality (xG).
- Liverpool mainly shoot from central penalty area → focus on high-quality chances.
- Rolling xG difference shows they outperformed opponents in 9 of 10 matches (except vs Man City).

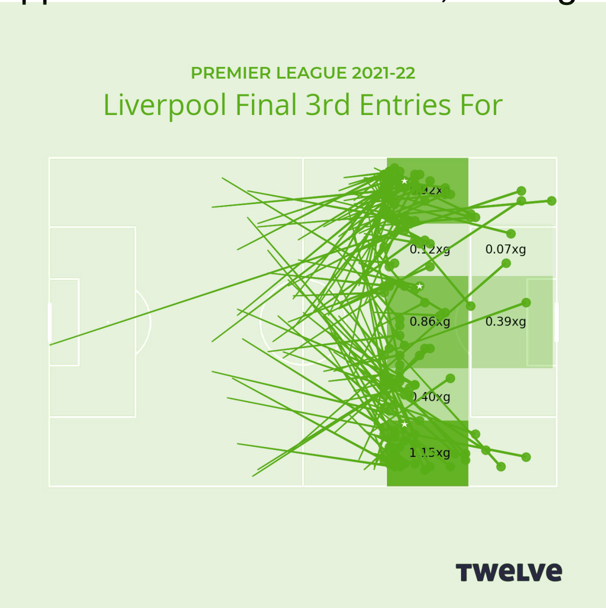


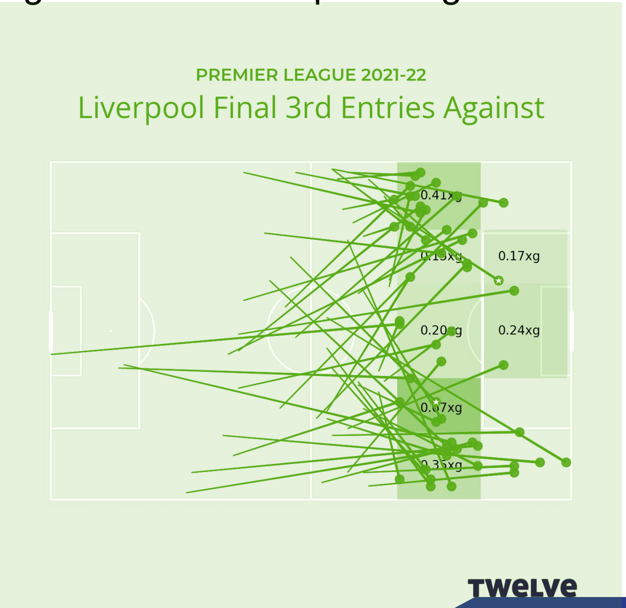


Key Performance Indices (KPI)

Final Third Entries

- Analysis of chance quality based on entry zones.
- Most dangerous attacks come from wide areas (wings).
- Central entries less threatening.
- Opponents enter less often, but slightly more dangerous from Liverpool's right side.





Key Performance Indices (KPI)

Transitions & Pressing

- Strong high press & ball regains leading to chances.
- More successful regains on left wing.
- Effective at converting regains in their own half into attacks.
- Opponents rarely create danger from regains against Liverpool.